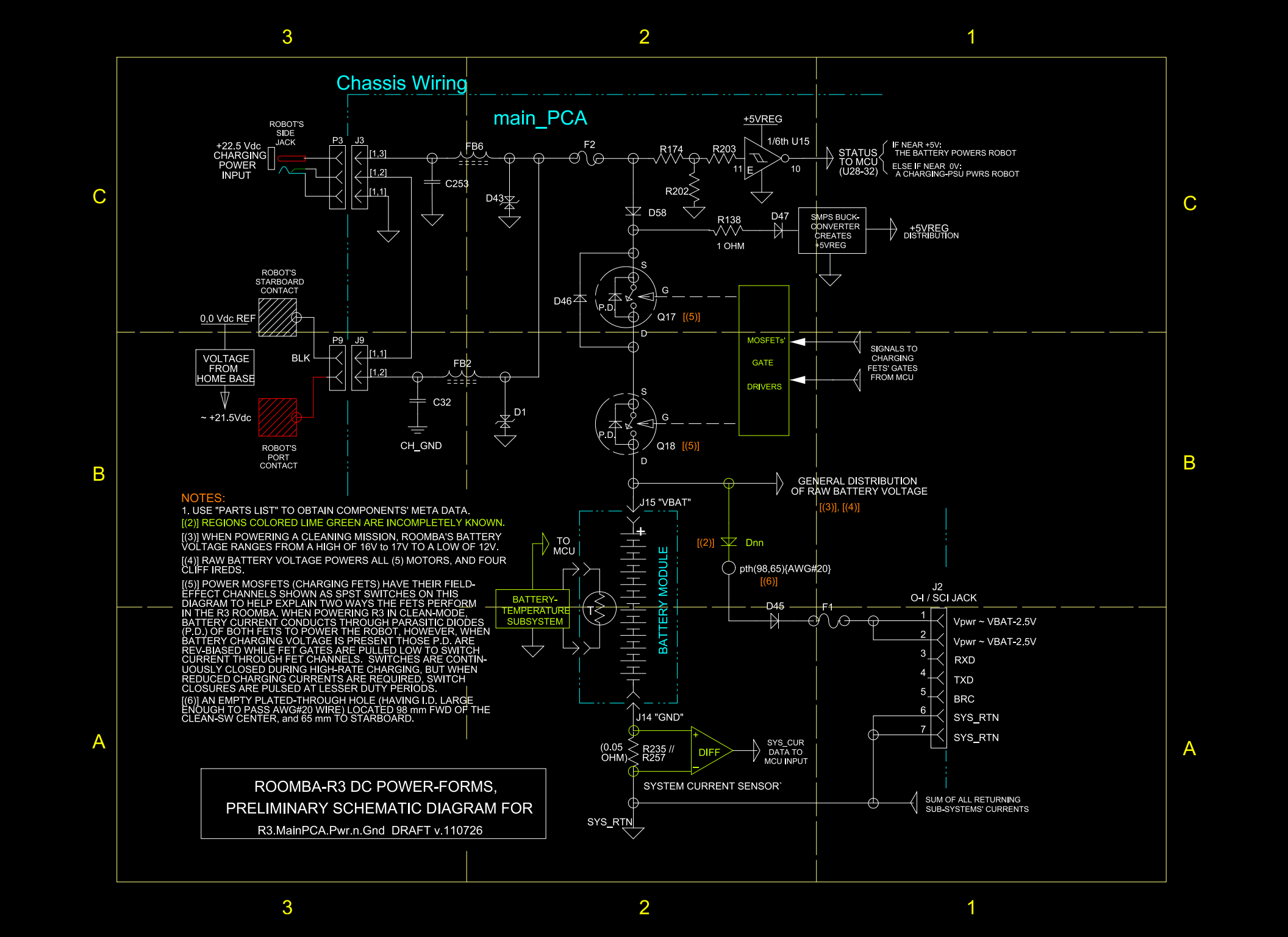

Chassis Wiring
main_PCA
ROBOT'S SIDE JACK
+22.5 Vdc CHARGING POWER INPUT
P3
J3
[1,3]
[1,2]
[1,1]
FB6
C253
D43
F2
R174
R203
R202
+5VREG
1/6th U15
11
E
10
STATUS TO MCU (U28-32)
IF NEAR +5V: THE BATTERY POWERS ROBOT
ELSE IF NEAR 0V: A CHARGING-PSU PWRS ROBOT
D58
R138
1 OHM
D47
SMPS BUCK-CONVERTER CREATES +5VREG
+5VREG DISTRIBUTION
ROBOT'S STARBOARD CONTACT
0,0 Vdc REF
VOLTAGE FROM HOME BASE
~ +21.5Vdc
ROBOT'S PORT CONTACT
P9
J9
BLK
[1,1]
[1,2]
FB2
C32
CH_GND
D1
D46
S
G
D
P.D.
Q17 [(5)]
Q18 [(5)]
MOSFETs' GATE DRIVERS
SIGNALS TO CHARGING FETS' GATES FROM MCU
GENERAL DISTRIBUTION OF RAW BATTERY VOLTAGE
[(3)], [(4)]
J15 "VBAT"
TO MCU
BATTERY-TEMPERATURE SUBSYSTEM
T
BATTERY MODULE
[(2)]
Dnn
pth(98,65){AWG#20}
[(6)]
D45
F1
J2
O-I / SCI JACK
1 Vpwr ~ VBAT-2.5V
2 Vpwr ~ VBAT-2.5V
3 RXD
4 TXD
5 BRC
6 SYS_RTN
7 SYS_RTN
J14 "GND"
(0.05 OHM)
R235 // R257
DIFF
SYS_CUR DATA TO MCU INPUT
SYSTEM CURRENT SENSOR`
SUM OF ALL RETURNING SUB-SYSTEMS' CURRENTS
SYS_RTN
NOTES:
1. USE "PARTS LIST" TO OBTAIN COMPONENTS' META DATA.
[(2)] REGIONS COLORED LIME GREEN ARE INCOMPLETELY KNOWN.
[(3)] WHEN POWERING A CLEANING MISSION, ROOMBA'S BATTERY VOLTAGE RANGES FROM A HIGH OF 16V to 17V TO A LOW OF 12V.
[(4)] RAW BATTERY VOLTAGE POWERS ALL (5) MOTORS, AND FOUR CLIFF IREDS.
[(5)] POWER MOSFETS (CHARGING FETS) HAVE THEIR FIELD-EFFECT CHANNELS SHOWN AS SPST SWITCHES ON THIS DIAGRAM TO HELP EXPLAIN TWO WAYS THE FETS PERFORM IN THE R3 ROOMBA. WHEN POWERING R3 IN CLEAN-MODE, BATTERY CURRENT CONDUCTS THROUGH PARASITIC DIODES (P.D.) OF BOTH FETS TO POWER THE ROBOT. HOWEVER, WHEN BATTERY CHARGING VOLTAGE IS PRESENT THOSE P.D. ARE REV-BIASED WHILE FET GATES ARE PULLED LOW TO SWITCH CURRENT THROUGH FET CHANNELS. SWITCHES ARE CONTIN-UOUSLY CLOSED DURING HIGH-RATE CHARGING, BUT WHEN REDUCED CHARGING CURRENTS ARE REQUIRED, SWITCH CLOSURES ARE PULSED AT LESSER DUTY PERIODS.
[(6)] AN EMPTY PLATED-THROUGH HOLE (HAVING I.D. LARGE ENOUGH TO PASS AWG#20 WIRE) LOCATED 98 mm FWD OF THE CLEAN-SW CENTER, and 65 mm TO STARBOARD.
ROOMBA-R3 DC POWER-FORMS, PRELIMINARY SCHEMATIC DIAGRAM FOR
R3.MainPCA.Pwr.n.Gnd DRAFT v.110726